Hypothesis Class Complexity and Learnability

Aaron Geelon So

The learning problem begins with *time*, or the lack thereof. Because both humans and machines have a finite amount of time to interact with the world and to perform computation, what is physically even possible to learn is also limited. In this paper, we will rely on topological and information-theoretic arguments to make a series of restrictions in the possible ways that a finite learner may model the world. Admittedly, the first few restrictions do not at this point lead to very many useful results. But importantly, they help show that the way we view the world is in fact a large byproduct of ourselves.

The Setting

In this paper, we will generally consider active learning for a binary classification problem. In the learning theory literature, $\mathcal{X}$ often denotes an *instance space*, or the space of all possible objects learning takes place on. A *concept* on $\mathcal{X}$ is then just a function

$$c : \mathcal{X} \to \{0, 1\}$$

sorting points into their respective classification. The learner usually maintains a collection of hypotheses $h : \mathcal{X} \to \{0, 1\}$ for c. The collection of these hypotheses, denoted $\mathcal{H}$ is therefore viewed as a function space over $\mathcal{X}$, so that $\mathcal{H} \subset \{0, 1\}^{\mathcal{X}}$.

Taking a step back, though, we can more generally think of $\mathcal{H}$ as a possible *model* for the world (with respect to the concept c), and individual hypotheses $h \in \mathcal{H}$ as possible *explanations*. One obvious form of learning can be done by obtaining labeled samples $(x_i, c(x_i))$, and then just ruling out inconsistent hypotheses. Intuitively, $\mathcal{H}$ is more *complex* if it is *a priori* able to explain more realizations of the world; however, a more powerful model necessarily takes more time to learn. In fact, even in the finite case, where $|\mathcal{H}| = C$, we know that to describe a single hypothesis takes $\lg C$ bits, growing with C.

We will now attempt to formalize this notion of *complexity* and its tradeoffs with *learnability*. In particular, we will show a series of possible reductions of model complexity that inevitably results from our finite ability to learn. Our first reduction will restrict $\mathcal{H}$ from arbitrary function spaces to certain continuous function spaces; this was inspired by the notion that all computable functions are continuous with respect to some topology, as described for example in [E04].

Learnable Models are Continuous: the world is a Cantor space

Let us imagine that the universe $\mathscr{U}$ a totally arbitrary object, an infinite source of possibility if you like. Furthermore, assume that every single interaction between the learner and the universe takes a unit of time, regardless of whether that interaction is taking a measurement or performing a computation. In this way, we can establish an equivalence between time and information—potentially, each unit of time can give way to one bit of new information about the universe. That is, each $t \in \mathbb{N}$ may be associated to one bit $b_t \in \{0, 1\}$.

To the learner, the *state* of the universe corresponds to some unknown point x^* in:

$$\mathcal{X} = \{0, 1\}^{\mathbb{N}},$$

where each coordinate represents the output of some observation or computation. However, because the

learner has finite capacity, they cannot determine precisely x^* itself. Instead, at each instance, the learner can knows only finitely many of the coordinates $i_1, \ldots, i_T$.

The collection of all points that could possibly model the universe to that learner at that time is of the form:

$$B := B(b_{i_1}, \ldots, b_{i_T}) = \{x \in \mathcal{X} : x_{i_t} = b_{i_t}\}.$$

These are the points in the universe are *indistinguishable* or *inseparable* relative to that learner at that moment in time. In particular, the learner should be able to determine in finite time whether a point x is contained in B. Importantly, set membership should be semidecidable. More generally, suppose we have some collection of $U_i \subset \mathcal{X}$ such that

$$\text{MEMBERSHIP}(x, U_i)$$

is semidecidable for each U_i (naturally, $i \in \mathbb{N}$); let's call it *open*. Then, it follows that:

1. $\text{MEMBERSHIP}(x, U_i \cap U_j)$ is semidecidable, for if x is contained in $U_i \cap U_j$, then just by checking for membership both in series will terminate in finite time.
2. Let $U = \bigcup_{i=1}^{\infty} U_i$. Then, $\text{MEMBERSHIP}(x, U)$ is semidecidable; it is possible to come up with an algorithm that would halt in finite time if x were not contained in some U_i (however, I won't prove it here as the main focus of this paper isn't on this first reduction) .

And of course, $\text{MEMBERSHIP}(x, \mathcal{X})$ is decidably true, and $\text{MEMBERSHIP}(x, \emptyset)$ is decidably false. Thus, the punchline is that collection $\mathcal{T}$ of subsets of $\mathcal{X}$ that a learner may computably determine set membership of forms a topology: closed under finite intersection and countable unions.

Let's state this in more traditional language within learning theory: returning to the notion that hypotheses are mappings

$$h : \mathcal{X} \to \{0, 1\},$$

then h is learnable in finite (but arbitrary long) time if and only if membership in $h^{-1}(\{0\})$ and $h^{-1}(\{1\})$ are semidecidable. In other words, h is a continuous function with respect to the above topology $\mathcal{T}$ on $\mathcal{X}$. Notice that the basis sets B above are in fact *cylinder sets* as defined in topology; the topology $\mathcal{T}$ is precisely the product topology on $\{0, 1\}^{\mathbb{N}}$, and in fact this shows that $\mathcal{X}$ is a Cantor space.

In summary, we have reduced the collection of all learnable concepts to the collection of all continuous binary function whose domain is a Cantor space. Unfortunately, this does not restrict the class of possible learnable functions much,[1] for although each concept may be learned in finite time, that time is not restricted in any way (indeed, continuous functions may be very irregular).

Our next reduction will be slightly more powerful, and will rely on information theory.

Learnable Models Have Finite VC Dimension

In the previous section, although we said that the learner only has a finite amount of time, we did not bound that time explicitly, nor did we try to understand how the our model complexity is allowed to grow as we increase the amount of time we have to learn (or equivalently, as we boost our computational power). In this section, we describe a framework introduced in the recent paper [KLMY17] that might allow us to make some headway into this problem.

Now, suppose that the amount of time we have is $O(m)$. Then, our universe corresponds to an approximation of $\{0, 1\}^{\mathbb{N}}$ (or more topologically, a quotient space of $\{0, 1\}^{\mathbb{N}}$). In particular, the learner is theoretically able to distinguish at most on the order of $n = 2^{O(m)}$ objects. It follows that the number of possible concepts to learn are on the order of $2^n = 2^{2^{O(m)}}$, so *a priori*, to even describe a single concept requires $n = 2^{O(m)}$ units of time. If the learner is human, they may be long dead before that happens.

[1] In fact, it follows that we can extend this result that any actually our universe is any continuous image of the Cantor space, and our labels may also be any continuous image of a Cantor space. In particular, $[0, 1]$ is one such space. The collection of all continuous functions from $[0, 1]$ to $[0, 1]$ is huge.

This suggests two simplifying conditions: (1) our model of the universe better be describable in at most polylog n time if we have any hope of learning, and (2) perhaps it is unreasonable to even require that we learn a concept on all of $\mathcal{X}$, which is size n—maybe it is both acceptable and useful to consider learning on a sample S of objects from $\mathcal{X}$.

Let us formalize this intuition according to [KLMY17].

Distributed Learning: a generalization

The problem that [KLMY17] considers is of the communication complexity of distributed learning, borrowing the concept of a communication protocol by [Y79]. We let Alice and Bob be two learners with a shared hypothesis class $\mathcal{H}$. The inputs for each learner is a collection of labeled sample points, S_A and S_B. We can think of these sample points as representing the information about the world that each respectively knows prior to communication. To communicate, they are allowed to send each other samples—in correspondence with communication protocols, each sample is worth one bit of information. The usual *communication complexity* or number of bits Alice and Bob need to send back and forth now corresponds to the *sample complexity*, the number of samples they need to send to each other.

This framework generalizes the traditional learning model where the learner obtains samples from either the world or a teacher in order to produce a best hypothesis (e.g. an empirical risk minimizer) to approximate the target concept c, while attempting to minimize the sample complexity. In particular, this consists of the the following setup:

- Alice is a teacher whose sample S_A is everything (i.e. in many settings, all samples $(x, c(x))$, though we may generalize).
- Bob is a learner with no prior information about c.
- The specific problem Alice and Bob are trying to solve: how can both Alice and Bob individually produce a good approximator $h \in \mathcal{H}$ of c while spending the least amount of communication? (In this case, how can Alice minimize one-way communication).

However, there are other natural learning settings that may fit under this framework. For example, we may have a learning setting where a teacher again wants to send the fewest number of samples possible to the learner. However, the teacher is unaware of the learner's hypothesis class. In this setting,

- The shared hypothesis class may correspond to every possible classification $\{0,1\}^{\mathcal{X}}$.
- Alice's sample S_A is everything as before.
- We may model Bob having a smaller hypothesis class by a collection of samples S_B that restricts the possible remaining hypotheses to that subset $\mathcal{H} \subset \{0,1\}^{\mathcal{X}}$.
- In the obvious setting, Bob should be able to exhibit a good approximation of c at the end of communication. But we may also consider the setting where his samples S_B are inconsistent with the true labels that Alice has; he should be able to learn that his model was too simplistic.

And finally, and perhaps the most naturally, given our discussion in the previous setting, both Alice and Bob play the roles of teacher and learner, each having different information to offer. Then, they face the joint learning problem: at the end of communication, can they both exhibit hypotheses that is consistent with all labeled data $S = S_A \cup S_B$? Or can they determine that in fact their hypothesis class was too simplistic, so no hypothesis is consistent with all of S.

This last point reminisces of the previous discussion on continuous hypotheses; indeed, we will focus on precisely that decision problem. A bit more formally: given two learners with samples S_A and S_B and a shared hypothesis class $\mathcal{H}$, is $\mathcal{H}$ *realizable* on the joint sample $S = S_A \cup S_B$ (in the sense that there exists some $h \in \mathcal{H}$ consistent with S)? Additionally, can we decide this in polynomial time?

This leads us to the following definitions of *complexity classes* for hypothesis classes. Here, we let $T : \mathbb{N} \to \mathbb{N}$ be any function. The function T is in terms of the sample size $|S| = n$, where S is the joint sample $S_A \cup S_B$. Succinctly, the class P corresponds to hypothesis classes where realizability of a sample of size n is decidable in a deterministic protocol in polylog n time.

Definition 1 (The class P)**.** *The* deterministic sample complexity $D_{\mathcal{H}}(n)$ *of* $\mathcal{H}$ *is the minimum number of samples communicated in a protocol to determines if* $\mathcal{H}$ *is realizable on an arbitrary sample of size* n*. If* $D_{\mathcal{H}}(n)$ *is bounded by some* $T(n)$*, we say that* $\mathcal{H}$ *has sample complexity bounded by* T*. When* $T(n) \leq \mathrm{poly}(\log n)$*, then we say that* $\mathcal{H}$ *is contained in* P*.*

Similarly, we can define the analogous non-deterministic sample complexity as the minimum subsample of S that can prove (non)realizability. Formally, let $\mathcal{Z} = \mathcal{X} \times \{0, 1\}$ be the space of all possible labeled examples. Let $\mathcal{Z}^*$ denote all possible finite samples of $\mathcal{X} \times \{0, 1\}$ (thus, $\mathcal{Z}^*$ is a disjoint union of $\mathcal{Z}^n$ over all finite n).

Definition 2 (The class NP)**.** *We say that the* non-deterministic sample complexity $N_{\mathcal{H}}(n)$ *is bounded by* T *if there exists predicates* $A, B : \mathcal{Z} \times \mathcal{Z} \to \{\mathrm{T}, \mathrm{F}\}$ *such that*

1. *For every realizable sample* $S = S_A \cup S_B \in \mathcal{Z}^n$*, there exists a proof* $P \in S^{T(n)}$ *such that* $A(S_A, P) = B(S_B, P) = \mathrm{T}$*.*
2. *For every non-realizable sample* $S = S_A \cup S_B \in \mathcal{Z}^n$*, and for any 'proof'* $P \in \mathcal{Z}^{T(n)}$*, either* $A(S_A, P) = \mathrm{F}$ *or* $B(S_B, P) = \mathrm{F}$*.*

If $N_{\mathcal{H}}(n) \leq \mathrm{poly}(\log n)$*, then we say that* $\mathcal{H}$ *is contained in* NP*.*

Notice that there is a trivial upper bound on $N_{\mathcal{H}}(n)$ of n. That is, if Alice and Bob jointly have n samples, they can just send each other their samples, incurring a total cost of n samples. And from there, they can certainly solve the realizability problem. But as mentioned before, n is too high of a cost.

This helps formalize the intuition that we want our hypothesis class or model of the world to be describable with less than $\mathrm{poly}(\log n)$ time: in the spirit of the scientific method, an NP hypothesis class is computationally falsifiable. And if we care about being able to *explain* the world (and not just predicting it), then a computationally falsifiable hypothesis class is certainly a prerequisite.[2]

Let's consider now how the additional requirement that $\mathcal{H}$ is NP might restrict the hypothesis class. The following is a proof given in [KLMY17] that shows that an NP hypothesis class must have finite VC dimension. This proof is actually quite simple, but it introduces a case where it is possible to turn this learning theory problem into an set disjointness problem, solved by information theory.

NP implies Finite VC Dimension

VC dimension is a commonly used method within learning theory to restrict the size of the hypothesis class. It requires the concept of *shattering* a space:

Definition 3 (Shattering)**.** *Let* $\mathcal{H} \subset \{0, 1\}^{\mathcal{X}}$ *be a hypothesis class over an instance space* $\mathcal{X}$*. We say that* $\mathcal{H}$ shatters *a subset* $S \subset \mathcal{X}$ *if the restriction of all hypotheses to* S *is* $\{0, 1\}^S$*. That is,*

$$\left\{h\big|_S : h \in \mathcal{H}\right\} = \{0, 1\}^S.$$

Essentially, $\mathcal{H}$ shatters S if there always exists some consistent hypothesis $h \in \mathcal{H}$ to any classification of points in S. We now define VC dimension:

Definition 4 (VC Dimension)**.** *The* VC dimension *of a hypothesis class* $\mathcal{H}$ *is the size of the largest set* $A \subset \mathcal{X}$ *that is shattered by* $\mathcal{H}$*.*

[2] Of course, somewhat disregarding the philosophical issue of why the world should be explainable at all—that most of the times, it doesn't surprise us too much. Perhaps our ability to explain reflects more of what and how we see rather than what the world in actuality is. In other words, we gloss over things that we can't explain and attribute it to randomness. In the sense of the first section, randomness might just be discontinuity.

Note that VC dimension may be infinite. However, it turns out that hypothesis classes with infinite VC dimension are not NP, according to this theorem:

Theorem 5 ("VC dim $\leq$ NP")**.** *For every class $\mathcal{H}$ with VC dimension $d \in \mathbb{N} \cup \{\infty\}$,*

$$N_{\mathcal{H}}(n) = \tilde{\Omega}(\min(d, n)),$$

where the tilde hides logarithmic factors.

This theorem is given as [Theorem 6, KLMY17]. First, let's show our desired corollary:

Corollary 6 (NP $\implies$ VC)**.** *If $\mathcal{H}$ is* NP*, then it has finite VC dimension d.*

Proof. Since $N_{\mathcal{H}}(n)$ is trivially bounded above by n, we have a sequence:

$$\tilde{\Omega}(\min d, n) \leq N_{\mathcal{H}}(n) \leq \text{poly}(\log n).$$

But for sufficiently large n, it is not the case that $\tilde{\Omega}(n) \leq \text{poly}(\log n)$. Instead, we must have $d < \infty$. □

Of course, given the definition of VC dimension, this is not a very surprising result in retrospect: if $\mathcal{H}$ shatters a set A, then to learn on that set, there is no hope but to label every single point of A individually. In other words, restricted to A, $\mathcal{H}$ represents the most complex hypothesis class possible, and we already knew that this is a hard learning problem.

But let's take a look at the proof of Theorem 5, for it reduces the problem to non-deterministic set disjointness from information theory for a proof. Given this general model of learning as a communication protocol, there may be more instances where we will rely on such a reduction.

Proof of Theorem 5. In general, the set-disjointness problem DISJ_n on $[n]$ induces a corresponding realizability problem. In particular, we may select n elements $\{x_1, \ldots, x_n\}$ from $\mathcal{X}$ to represent each one of the n objects in the set-disjointness problem.

In DISJ_n, Alice and Bob are given subset $A, B \subset [n]$. In the realizability problem, we can then construct the corresponding input samples S_A and S_B, where Alice's points are all positively labeled and Bob's points are negatively labeled.

While in this general form, DISJ_n and realizability are not equivalent. However, in the case that $\mathcal{H}$ shatters a collection of points $R = \{x_1, \ldots, x_d\}$, then, a sample $S_A \cup S_B$ contained in R is nonrealizable if and only if there is some point with both positive and negative label:

$$(x_i, +) \in S_A \quad \text{and} \quad (x_i, -) \in S_B.$$

So given a hypothesis class $\mathcal{H}$ with VC dimension d, we can use the realizability protocol to solve specifically the disjointness problem on d points, DISJ_d. This way, we obtain a lower bound from the non-deterministic disjointness problem—a proof of (non)realizability using k samples translates to a proof of (non)disjointness using $k \lg d$ bits. And since this is bounded below by $\Omega(d)$, the number of samples required is $k = \Omega(d/\lg d)$. And of course, if $n < d$, then $\mathcal{H}$ shatters n points in $\mathcal{X}$, so the analogous statement holds for n when $n < d$. □

Sample Complexity and Packaged Information

Perhaps we should take care to examine the lower bound $\Omega(d/\lg d)$, for this seems to suggest that we might be able to solve set-disjointness with less than d communication cost. But recall that there is a bit of difference between a usual communication protocol and this distributed-learning protocol. In the latter, the atomic unit of communication are pairs (x, y) of labeled samples, containing more than one bit of information. How many bits of information are contained in (x, y)? If $|\mathcal{X}| = O(n)$, then at most, (x, y) contains $O(\lg n)$ bits of information, corresponding to the log factor, $1/\lg d$, in the above lower bound.

It follows that any distributed-learning protocol may be simulated by a usual communication protocol where the communication complexity is related to the sample complexity by a blowup factor of $O(\lg n)$. Conversely,

any communication protocol may be simulated by a distributed-learning protocol—instead of sending bits 0 and 1, Alice and Bob may establish the meaning of certain sample points to represent bits of information. Importantly, a poly($\log n$)-time algorithm expanded by a factor of $\lg n$ is still a poly($\log n$)-time algorithm.

And so, if we fundamentally care about computational efficiency—polylogarithmic in time/bits or equivalently, polylogarithmic in samples—then this distributed model of learning potentially provides a useful and flexible connection to communication complexity in general. Let me conclude with a few possible directions to consider.

Further Research

In order of increasing undefinedness:

1. **Other modes of learning:** Can we concretely establish a way of translating other formulation of learning to this communication-based model of learning?

2. **Priors:** when learning proceeds by searching for a consistent hypothesis, we model learning by reducing the size of $\mathcal{H}$ combinatorially. But we can also say the same thing in a distributional setting (analogous to how entropy connects probability distributions to with combinatorics). That is, the combinatorial reduction is just the restriction of the support on $\mathcal{H}$. So, in general, the goal of learning is perhaps not looking for a consistent hypothesis, but to reduce the entropy of $\mathcal{H}$. The distributed-learning setup also allows for this generalization. It is currently formulated deterministically, but there are many reasonable ways we can add distributions to the model.

3. **NP beyond realizability:** The NP class of hypotheses here was formulated specifically for the realizability decision problem. And more generally, the distributed-learning model in the original paper was not as highly motivated by computation as presented here. Perhaps there are natural ways to extend this to be able to answer broader questions about, say, the impossibility of learning certain models. Or perhaps more usefully, understanding what conditions computational efficiency will induce on the hypothesis class. As a result, we may be able to obtain stronger results in learning theory given more (reasonable) conditions.

4. **Multi-part distributed learning and Privacy:** The extension to multi-party is an obvious direction. What is also interesting is a question of privacy. Suppose each person has a bunch of private information they want to contribute to the learning model; however, we want to ensure that it is impossible to deduce the source of that information. Are there ways we can guarantee privacy? What's the least amount of information I can send to someone else or make public so that collectively we can learn, but also maintain privacy on the rest of my data? (Assuming perhaps there is some correlation between the subset of information that I know). As a concrete example, perhaps we want to train a neural network on personal data.

5. **Randomness and pseudorandomness:** the notion of *separability* is fundamental to topology; in fact, there are a whole bunch of separation axioms (e.g. common conditions include T_0, Hausdorff, etc.)—the driving intuition is that open sets essentially encode equivalence. Points that have 'many' shared open sets are harder to distinguish. In the trivial case where any open set containing x must also contain y, then any continuous functions must act on y the same way it acts on x.

   Generally, continuous functions therefore represent measurements, interactions, or actions that respect that equivalence. We can model a gain in discerning power by a filtration of topologies $\mathcal{T}_1 \subset \mathcal{T}_1 \subset \cdots$, so as a result, functions that had appeared discontinuous to us before can now look continuous. In other words, we have greater ability to explain the world. And on the other hand, discontinuity seems random to us, because they symmetry is broken, seemingly at random.

   This, then, is one way we can try to think about randomness and information: if we are restricted to a certain finite class of observables, then many continuous functions on the Cantor space will appear discontinuous to us on the topology induced by the observables—we might call such functions

pseudorandom. Generalizing, perhaps 'true randomness' are functions that are discontinuous even with respect to the topology on the Cantor set.

References

[BBCR13] Barak, B., Braverman, M., Chen, X., Rao, A. "How to compress interactive communication." SIAM Journal on Computing 42.3 (2013): 1327-1363.

[B15] Braverman, M. "Interactive information complexity." SIAM Journal on Computing 44.6 (2015): 1698-1739.

[BHLC16] Beygelzimer, A., Hsu, D., Langford J., Zhang, C. "Search improves label for active learning." Advances in Neural Information Processing Systems. 2016.

[CAL94] Cohn, D., Atlas, L., Ladner, R. "Improving generalization with active learning." Machine Learning, 15(2):201–221, 1994.

[E04] Escardó, Martín. "Synthetic topology: of data types and classical spaces." Electronic Notes in Theoretical Computer Science 87 (2004): 21-156.

[H10] Hsu, D. "Algorithms for active learning." (2010). Learning 2010.

[KLMY17] Kane, Daniel M., et al. "On Communication Complexity of Classification Problems." arXiv preprint arXiv:1711.05893 (2017)

[Y79] Yao, Andrew Chi-Chih. "Some complexity questions related to distributive computing." Proceedings of the eleventh annual ACM symposium on Theory of computing. ACM, 1979.